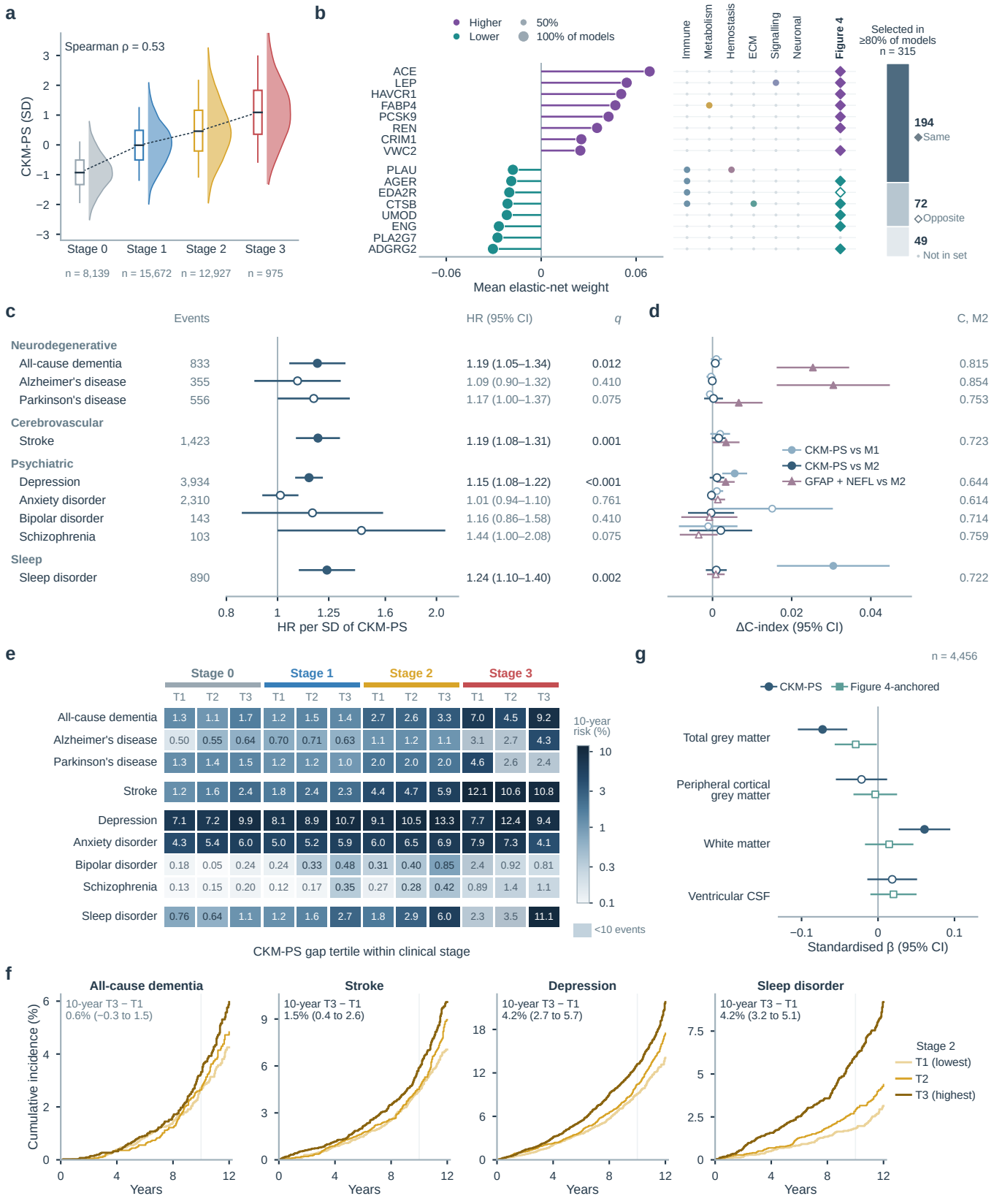

a
Spearman ρ = 0.53
CKM-PS (SD)
Stage 0
Stage 1
Stage 2
Stage 3
n = 8,139
n = 15,672
n = 12,927
n = 975
b
Higher
Lower
50%
100% of models
ACE
LEP
HAVCR1
FABP4
PCSK9
REN
CRIM1
VWC2
PLAU
AGER
EDA2R
CTSB
UMOD
ENG
PLA2G7
ADGRG2
Mean elastic-net weight
Immune
Metabolism
Hemostasis
ECM
Signalling
Neuronal
Figure 4
Selected in ≥80% of models n = 315
194 Same
72 Opposite
49 Not in set
c
Events
HR (95% CI)
q
Neurodegenerative
All-cause dementia 833 1.19 (1.05–1.34) 0.012
Alzheimer's disease 355 1.09 (0.90–1.32) 0.410
Parkinson's disease 556 1.17 (1.00–1.37) 0.075
Cerebrovascular
Stroke 1,423 1.19 (1.08–1.31) 0.001
Psychiatric
Depression 3,934 1.15 (1.08–1.22) <0.001
Anxiety disorder 2,310 1.01 (0.94–1.10) 0.761
Bipolar disorder 143 1.16 (0.86–1.58) 0.410
Schizophrenia 103 1.44 (1.00–2.08) 0.075
Sleep
Sleep disorder 890 1.24 (1.10–1.40) 0.002
HR per SD of CKM-PS
d
C, M2
CKM-PS vs M1
CKM-PS vs M2
GFAP + NEFL vs M2
0.815
0.854
0.753
0.723
0.644
0.614
0.714
0.759
0.722
ΔC-index (95% CI)
e
Stage 0
Stage 1
Stage 2
Stage 3
T1 T2 T3 T1 T2 T3 T1 T2 T3 T1 T2 T3
All-cause dementia 1.3 1.1 1.7 1.2 1.5 1.4 2.7 2.6 3.3 7.0 4.5 9.2
Alzheimer's disease 0.50 0.55 0.64 0.70 0.71 0.63 1.1 1.2 1.1 3.1 2.7 4.3
Parkinson's disease 1.3 1.4 1.5 1.2 1.2 1.0 2.0 2.0 2.0 4.6 2.6 2.4
Stroke 1.2 1.6 2.4 1.8 2.4 2.3 4.4 4.7 5.9 12.1 10.6 10.8
Depression 7.1 7.2 9.9 8.1 8.9 10.7 9.1 10.5 13.3 7.7 12.4 9.4
Anxiety disorder 4.3 5.4 6.0 5.0 5.2 5.9 6.0 6.5 6.9 7.9 7.3 4.1
Bipolar disorder 0.18 0.05 0.24 0.24 0.33 0.48 0.31 0.40 0.85 2.4 0.92 0.81
Schizophrenia 0.13 0.15 0.20 0.12 0.17 0.35 0.27 0.28 0.42 0.89 1.4 1.1
Sleep disorder 0.76 0.64 1.1 1.2 1.6 2.7 1.8 2.9 6.0 2.3 3.5 11.1
10-year risk (%)
10
3
1
0.3
0.1
<10 events
CKM-PS gap tertile within clinical stage
g
n = 4,456
CKM-PS
Figure 4-anchored
Total grey matter
Peripheral cortical grey matter
White matter
Ventricular CSF
Standardised β (95% CI)
f
All-cause dementia
10-year T3 − T1
0.6% (−0.3 to 1.5)
Stroke
10-year T3 − T1
1.5% (0.4 to 2.6)
Depression
10-year T3 − T1
4.2% (2.7 to 5.7)
Sleep disorder
10-year T3 − T1
4.2% (3.2 to 5.1)
Cumulative incidence (%)
Years
Stage 2
T1 (lowest)
T2
T3 (highest)